

AI Speaking Assistant

Use-Case Specification: UC5.3 Evaluate Performance on Specific Criteria

Version: 1.0

Revision History

Date	Version	Description	Author
23/Jul/26	1.0	Initial RUP specification draft for UC5.3	Gia Phúc

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1. Use-Case Name

UC5.3: Evaluate performance on specific criteria

1.1 Brief Description

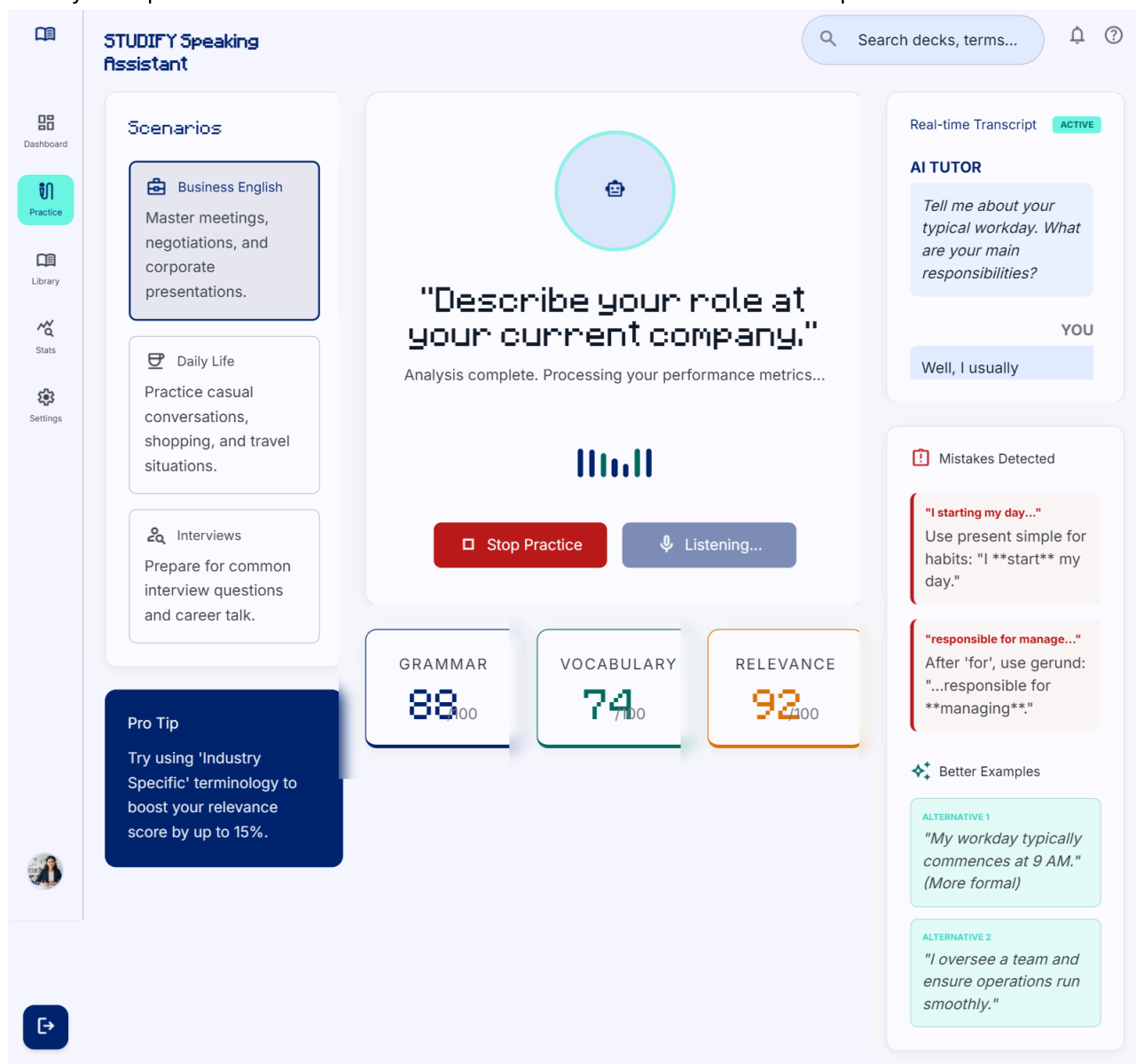
This use case details how the system analyzes the Learner's transcribed spoken speech against specific core linguistic criteria (Grammar, Vocabulary & Word Choice, and Context Relevance) to produce structured analytical scores and diagnostic feedback.

2. Flow of Events

2.1 Basic Flow

1. The system initiates a performance evaluation routine upon turn completion.
2. The system executes **UC5.1: Speech recognition** to capture/verify the input text (<<include>>).
3. The system submits the transcript and active target evaluation rubric to the AI Engine.
4. The AI Engine evaluates performance across three strict criteria:
 - **Grammar & Syntax:** Identifies structural, tense, and agreement errors.
 - **Vocabulary & Word Choice:** Assesses lexical variety, appropriateness, and sophistication.
 - **Context Relevance:** Evaluates how accurately the utterance addresses the scenario prompt.
5. The system records numerical scores and qualitative diagnostic notes.

6. The system presents the detailed breakdown on the interface evaluation panel.



2.2 Alternative Flows

2.2.1 Brief Input Evaluation

1. In step 4 of the Basic Flow, the AI Engine determines that the user utterance is too brief (e.g., single-word responses like "Yes" or "Okay") to perform full grammatical analysis.
2. The AI Engine assigns neutral baseline scores and notes: *"Utterance too brief for detailed grammatical breakdown."*

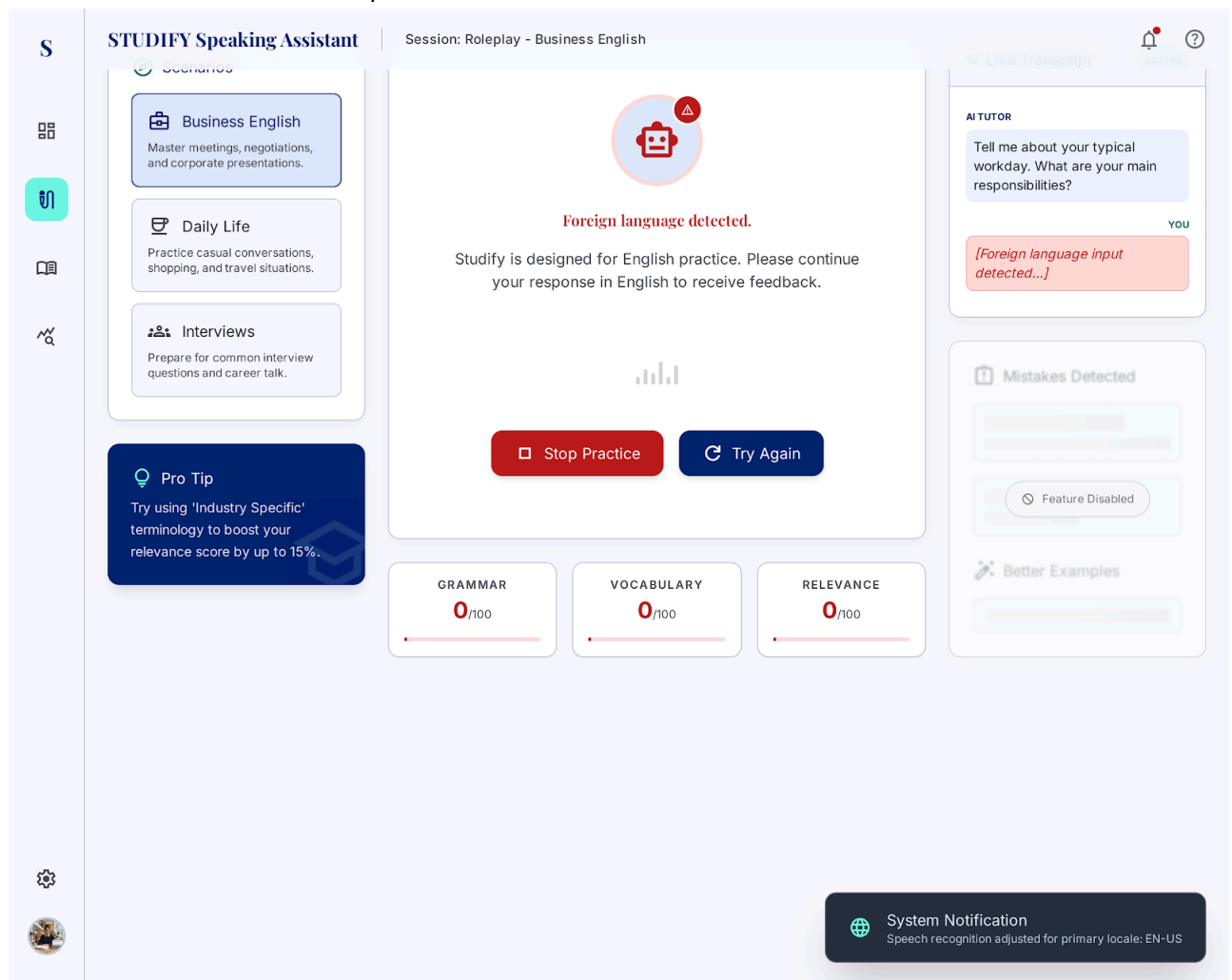
3. The execution resumes at step 5 of the Basic Flow.

The screenshot displays the STUDIFY Speaking Assistant interface. On the left, a sidebar contains icons for a home screen, a list of scenarios, a book, and a search icon. The main area is titled "STUDIFY Speaking Assistant" and features a search bar at the top right. Below the title, there are three scenario cards: "Business English" (Master meetings, negotiations, and corporate presentations), "Daily Life" (Practice casual conversations, shopping, and travel situations), and "Interviews" (Prepare for common interview questions and career talk). A "Pro Tip" box suggests using 'Industry Specific' terminology to boost relevance scores. The central panel shows a listening exercise with the prompt "Describe your role at your current company." and a feedback message: "Utterance too brief for detailed grammatical breakdown." Below this are "Stop Practice" and "Listening..." buttons. At the bottom, three progress bars show scores for Grammar (50/100), Vocabulary (50/100), and Relevance (50/100). On the right, a "Real-time Transcript" section shows the AI tutor's prompt and the user's response "Yes." Below this, a "Mistakes Detected" section shows "N/A" with a message: "No mistakes detected in brief input. Continue speaking to get feedback." A "BETTER EXAMPLES" section provides a more detailed response: "Yes, I'd be happy to. At my current job, I focus primarily on..."

2.2.2 Non-English Speech Input Detected

1. In step 4 of the Basic Flow, the AI Engine detects that the transcribed text is in a language other than English.
2. The system assigns a score of 0 for Vocabulary and Grammar, flagging the turn with: *"Foreign language detected. Please practice in English."*

3. The execution resumes at step 5 of the Basic Flow.



2.2.3 Rubric Configuration Missing

1. In step 3 of the Basic Flow, the system fails to load target scenario rubric configuration parameters.
2. The system falls back to a standardized default evaluation rubric and logs a system warning.

3. The execution resumes at step 4 of the Basic Flow.

The screenshot displays the STUDIFY Speaking Assistant interface during a listening practice session. The interface is divided into several sections:

- Header:** Includes the STUDIFY logo, "Speaking Assistant", "Standard Assessment Mode", and "Academic Session".
- Scenarios:** A list of practice scenarios on the left:
 - Business English:** Master business meetings, negotiations, and corporate presentations.
 - Daily Life:** Practice casual conversations, shopping, and travel situations.
 - Interviews:** Prepare for common interview questions and career talk.
 - Pro Tip:** Try using 'Industry Specific' terminology to boost your relevance score by up to 15%.
- Central Practice Area:**
 - A large blue circle with a white dot in the center.
 - Text prompt: "Describe your role at your current company." followed by "I'm listening... Speak clearly for the best transcription."
 - A small bar chart showing three bars of increasing height.
 - Buttons: "Stop Practice" (red) and "Listening..." (blue).
- Scorecards:** At the bottom, three scorecards are displayed:
 - GRAMMAR:** 88/100
 - VOCABULARY:** 74/100
 - RELEVANCE:** 92/100
- REAL-TIME TRANSCRIPT:** On the right, a section titled "REAL-TIME TRANSCRIPT" (ACTIVE) shows the AI Tutor's prompts and the user's response.
 - AI TUTOR: "Tell me about your typical workday. What are your main responsibilities?"
 - YOU: "Well, I usually starting my day at 9 AM. I am responsible for manage the team."
 - AI TUTOR: "That sounds interesting. How do you plan your..."
- MISTAKES DETECTED:** A section below the transcript highlighting errors:
 - "I starting my day..." - Use present simple for habits: "I *start* my day."
 - "responsible for manage..." - After 'for', use gerund: "...responsible for *managing*."
- BETTER EXAMPLES:** A section at the bottom right providing alternative phrasings:
 - ALTERNATIVE 1:** "My workday typically commences at 9 AM." (More formal)
 - ALTERNATIVE 2:** "I oversee a team and ensure operations run smoothly."

3. Special Requirements

3.1 Multi-Criteria Scoring Consistency

The evaluation module must generate standardized scores (0–100 scale) along with human-readable diagnostic rationales for Grammar, Vocabulary, and Context Relevance.

3.2 Evaluation Latency

Evaluation calculations and metric scoring must be completed within 1.5 seconds following text input ingestion.

3.3 Deterministic Scoring Standards

Scoring guidelines must adhere to standardized CEFR (Common European Framework of Reference for Languages) proficiency benchmarks.

4. Preconditions

4.1 Transcribed Input Available

The Learner's spoken input must be successfully processed into text format.

4.2 Evaluation Rubric Loaded

Target scenario rubric criteria must be active in session memory.

4.3 Scenario Context Set

The specific practice domain (e.g., Job Interview, Restaurant Order) must be active to measure contextual relevance accurately.

5. Postconditions

5.1 Metrics Logged

Detailed evaluation scores and criterion breakdowns are logged in the active session report.

5.2 UI Performance Panel Updated

Visual score cards and radar charts on the learner's screen are updated with new metrics.

5.3 Analytics Trend Updated

Scores are appended to the learner's long-term progress profile database.

6. Extension Points

6.1 Performance Weakness Detected

- **Location:** Step 5 of the Basic Flow.
- **Condition:** If performance scores fall below set thresholds or specific actionable errors/weaknesses are flagged, the system invokes **UC5.4: Provide guidance on how to improve** (<<extend>>).